

Fall 2007

Problem 1 (Fa07). Let $\mathbb{Z}[i]$ be the ring of complex numbers of the form $a + bi$, $a, b \in \mathbb{Z}$. List all nonisomorphic subrings of $\mathbb{Z}[i]$.

Problem 2 (Fa07). Let f and g be entire functions such that $f'(z) = g(z)$, $g'(z) = -f(z)$, and $f(2z) = 2f(z)g(z)$ for all $z \in \mathbb{C}$. Find all possibilities for f .

Problem 3 (Fa07). Let A be an $n \times n$ Hermitian matrix, and $x \in \mathbb{C}^n$ be a vector such that $A^2x = 0$. Prove that $Ax = 0$.

Problem 4 (Fa07). Let (a_n) and (b_n) be sequences of real numbers. Suppose that $0 \leq a_{n+1} \leq a_n + b_n$ for all $n \geq 1$, and that $\sum_{n=1}^{\infty} b_n$ converges. Prove that $\lim_{n \rightarrow \infty} a_n$ exists and is finite.

Problem 5 (Fa07). Suppose that G is a finite group such that for each subgroup H of G there exists a homomorphism $\varphi: G \rightarrow H$ such that $\varphi(h) = h$ for all $h \in H$. Show that G is a product of groups of prime order.

Problem 6 (Su80, Fa83, Sp84, Fa87, Fa96, Fa07). How many zeros does the complex polynomial

$$3z^9 + 8z^6 + z^5 + 2z^3 + 1$$

have in the annulus $1 < |z| < 2$?

Problem 7 (Fa07). Let $P \in \mathbb{R}^{n \times n}$ be a matrix satisfying $P^3 = P$. Let r be the rank of P and assume $r > 0$. Show that there exist matrices $U, V \in \mathbb{R}^{n \times r}$ satisfying $V^T U = I_r$ such that

$$P = USV^T$$

where I_r is the $r \times r$ identity matrix, and S is an $r \times r$ diagonal matrix with ± 1 on the diagonal.

Problem 8 (Fa07). Suppose that (b_n) is a sequence of positive real numbers tending to infinity such that $b_n/n \rightarrow 0$. Must there exist a sequence (a_n) such that $(a_1 + \cdots + a_n)/n \rightarrow 0$ and $\limsup_{n \rightarrow \infty} (a_n/b_n) = \infty$?

Problem 9 (Fa07). Let G be a non-abelian group of order 16 having a subgroup H isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$. Prove that the number of elements of G of order 2 is either 7 or 11.

Problem 10 (Fa07). Let $f(z)$ be a polynomial with complex coefficients, and let a be a complex number. Prove that $\{a, f(a), f(f(a)), \dots\}$ is not dense in $\mathbb{C}$.

Problem 11 (Fa07). Let A be an $n \times n$ complex matrix. Suppose that m is a positive integer such that A^m is diagonalizable. Prove that A^{m+1} is diagonalizable.

Problem 12 (Fa07). Let $\{u_1, u_2, \dots, u_k\}$ be a set of linearly independent vectors in $\mathbb{R}^n$, and let $\mathcal{A}$ be a closed set in $\mathbb{R}^k$. Let S be the set of linear combinations $\alpha_1 u_1 + \alpha_2 u_2 + \dots + \alpha_k u_k$ obtained as $(\alpha_1, \alpha_2, \dots, \alpha_k)$ ranges over all points of $\mathcal{A}$. Show that S is a closed subset of $\mathbb{R}^n$.

Problem 13 (Fa07). Let $\mathbb{K}$ and $\mathbb{L}$ be fields, and let $\mathbb{K} \times \mathbb{L}$ be the product ring, with addition and multiplication defined componentwise. Find all prime ideals of $\mathbb{K} \times \mathbb{L}$.

Problem 14 (Fa07). Let f be an entire function and $a_1, \dots, a_n$ be all zeros of f in $\mathbb{C}$. Suppose that there exist real numbers $R > 0$ and $a > 1$ such that $|f(z)| \geq |z|^a$ for all $|z| \geq R$. Prove that

$$\sum_{j=1}^n \operatorname{Res} \left(\frac{1}{f(z)}, a_j \right) = 0$$

Problem 15 (Fa07). Given a positive integer n , what are the possible values of the triple $(\operatorname{rank} A, \operatorname{rank} B, \operatorname{rank} C)$ as A, B, C range over real $n \times n$ matrices satisfying $A + B + C = 0$?

Problem 16 (Fa07). Let $f : [0, \infty) \rightarrow \mathbb{R}$ be continuous. Suppose that $\lim_{x \rightarrow \infty} f(x)$ exists and is finite. Must f be uniformly continuous?

Problem 17 (Fa07). Show that for every positive integer n , there exists an irreducible polynomial over $\mathbb{Q}$ of degree n such that all its zeros are real.

Problem 18 (Fa07). Let f be holomorphic on a neighborhood of the closed disc $\overline{\mathbb{D}} = \{z \mid |z| \leq 1\}$. Suppose that $\max_{|z|=1} |f(z)| \leq 1$. Prove that there exists a complex number z such that $|z| \leq 1$ and $f(z) = z$.